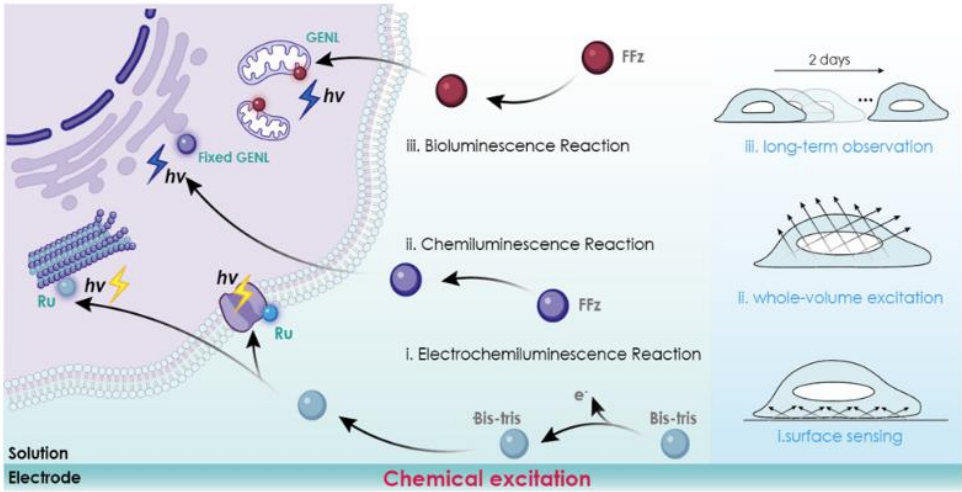
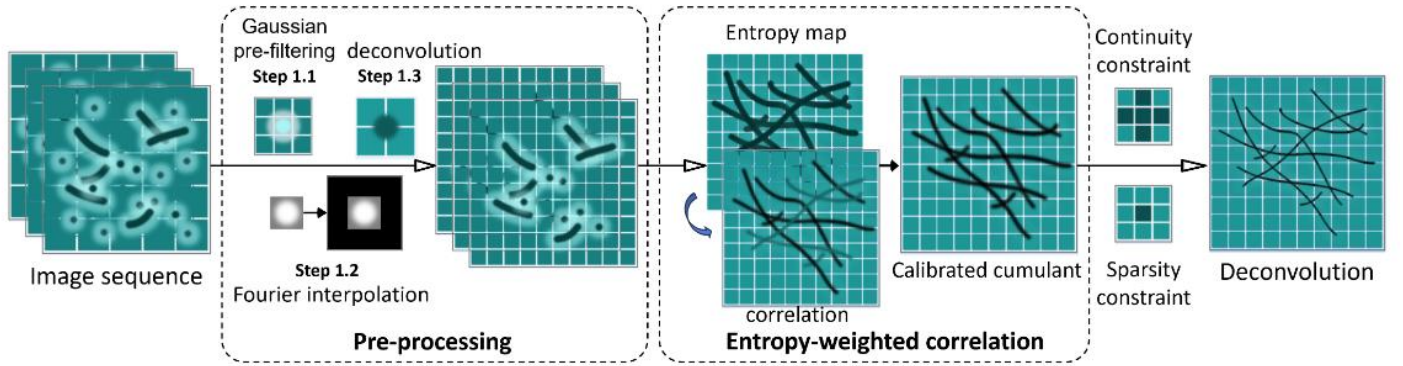


Introduction



The **RIED** is a new conceptual and methodological framework for super-resolution luminescence imaging. It establishes that fluctuation information inherent in reaction-driven luminescence—including **electrochemiluminescence (ECL)**, **chemiluminescence (CL)**, and **bioluminescence (BL)**—can be harnessed to achieve super-resolution reconstruction, enabling zero-background, ultra-long-term super-resolution imaging with sub-100 nm resolution. Within this framework, we developed a specific computational reconstruction algorithm, which implements entropy-weighted correlation with dual-step deconvolution to extract super-resolved information from continuously collected short-exposure frames.

RIED reconstruction workflow



Main step	Function
Pre-processing	Gaussian pre-filtering to suppress high-frequency sampling noise; Fourier interpolation to provide a finer grid and ensures sufficient pixel support for subsequent resolution enhancement; Pre-deconvolution to further suppress sampling noise and enhance fluctuation.
Entropy-weighted correlation	Identify emitters and extract super-resolved information
Post sparse deconvolution	Maximize resolution enhancement without introducing artefacts

Installation

• Tested platforms

Matlab 2022b, with Wavelet Toolbox, Image Processing Toolbox, Parallel Computing Toolbox (Win 10, 128 GB RAM, NVIDIA RTX 4090 24 GB, CUDA 11.6)

• Quick start

```
% Add RIEDm to your path
addpath(genpath('RIEDm'));

% Basic reconstruction
imgstack = imreadstack('bl.tif');
RIEDrecon = RIEDm(imgstack);
```

Parameters

• Basic parameters

Parameter	Description	Default	When to adjust
<code>pixel</code>	Pixel size (nm)	160	Match the microscope acquisition
<code>NA</code>	Numerical aperture	1.45	Match the objective lens
<code>wavelength</code>	Emission wavelength (nm)	620	Match the luminescence peak

• Advanced parameters

Parameter	Description	Recommended Range	Default	When to adjust
<code>gauss</code>	Gaussian pre-filter kernel size	0.5-2	0.7	Higher for low photon budget
<code>iter1</code>	Pre-deconvolution iterations	2-10	4	Increase to further enhance fluctuation and resolution, reduce if introduces artefacts
<code>subfactor</code>	Subtraction factor for cumulant	0-1	0.5	Increase if high fluctuation
<code>wavelet</code>	Weight for wavelet defocus signal estimation	0; 1-5	0	Set to '0' for ECL , where defocus signal is negligible. For CL and BL , lower values remove defocus signal more aggressively
<code>fidelity</code>	Fidelity weight for sparse deconvolution	10-100	50	Lower for high continuity constraint
<code>sparsity</code>	Sparsity weight for sparse deconvolution	0.1-5	0.5	Higher for further resolution improvement, Lower if real structure signals are filtered out
<code>finter2</code>	Factor of fourier interpolation after entropy-weighted correlation	1-2	1	'1' is enough for most cases . Increase if higher resolution demand.
<code>iter2</code>	Sparse deconvolution iterations	2-15	5	Increase to further enhance fluctuation, reduce if introduces artefacts
<code>gamma</code>	Factor of intensity correction	0.5-1	1	reduce to further intensity correction



Example demonstration

• Reconstruction for ECL

(microtubules, 100 frames)

```
% Involve RIED
addpath(genpath('./RIED_core'));
addpath(genpath('./Utils'));

% RIED recon
imgstack = imreadstack('ec11.tif');
RIEDrecon =
RIEDm(imgstack,'pixel',160,'NA',1.45,'wavelength',620,'iter1',3,'subfactor',0.5,'fidelity',50,'sparsity',3,'iter2',7,'gamma',0.8
);

% Visualization
baseline = 520;
Rawimg = imfilter(mean(imgstack,3),generate_rsf(1));
Rawimg = Rawimg-baseline;
Rawimg(Rawimg<0) = 0;
Rawimg = percennorm(Rawimg,1,100);
RIEDrecon = imfilter(RIEDrecon,generate_rsf(3));
RIEDrecon = percennorm(RIEDrecon,1,99.98);
load('colormap_green.mat')
figure,
subplot(1,2,1),imshow(Rawimg,[0 1],'colormap',colormap_green),title('Raw summed data')
subplot(1,2,2),imshow(RIEDrecon,[0 1],'colormap',colormap_green), title('RIED')
```

• Reconstruction for ECL

(microtubules, 50 frames)

```
% Involve RIED
addpath(genpath('./RIED_core'));
addpath(genpath('./Utils'));

% RIED recon
imgstack = imreadstack('ec12.tif');
RIEDrecon = RIEDm(imgstack,'pixel',160,'NA',1.45,'wavelength',620,'iter1',3,'subfactor',1,'fidelity',20,'sparsity',1,'iter2',3);

% Visualization
baseline = 580;
Rawimg = imfilter(mean(imgstack,3),generate_rsf(1));
Rawimg = Rawimg-baseline;
Rawimg(Rawimg<0) = 0;
Rawimg = percennorm(Rawimg,1,100);
RIEDrecon = imfilter(RIEDrecon,generate_rsf(1));
RIEDrecon = percennorm(RIEDrecon,1,99.5);
load('colormap_green.mat')
figure,
subplot(1,2,1),imshow(Rawimg,[0 1],'colormap',colormap_green),title('Raw summed data')
subplot(1,2,2),imshow(RIEDrecon,[0 1],'colormap',colormap_green), title('RIED')
```

• **Reconstruction for BL**

(mitochondria, 200 frames)

```
% Involve RIED
addpath(genpath('./RIED_core'));
addpath(genpath('./Utils'));

% RIED recon
imgstack = imreadstack('bl.tif');
RIEDrecon =
RIEDm(imgstack,'pixel',160,'NA',1.42,'wavelength',517,'iter1',4,'subfactor',0,'fidelity',150,'sparsity',20,'iter2',5);

% Visualization
baseline = 560;
Rawimg = imfilter(mean(imgstack,3),generate_rsf(2));
Rawimg = Rawimg-baseline;
Rawimg(Rawimg<0) = 0;
Rawimg = percennorm(Rawimg,1,100);
RIEDrecon = imfilter(RIEDrecon,generate_rsf(2));
RIEDrecon = percennorm(RIEDrecon,1,99.98);
load('colormap_yellowhot.mat')
figure,
subplot(1,2,1),imshow(Rawimg,[0 1],'colormap',colormap_yellowhot),title('Raw summed data')
subplot(1,2,2),imshow(RIEDrecon,[0 1],'colormap',colormap_yellowhot), title('RIED')
```

 Version

Version	Changes
v0.8.0	RIED reconstruction core for ECL, CL, and BL
v0.2.0	RIED reconstruction core for ECL
v0.1.0	initial version